



Determining Sample Size Transcript

Hello everyone, welcome to lesson 1 video 2. In this video, we'll see how to determine the sample size needed for a study. Choosing the right sample size is critical. Too small and the results are unreliable.

Too large, the resources are wasted. Let's see how statistics provide a systematic way to balance precision and efficiency. The main topic in this video includes margin of error, confidence level and confidence interval.

Hi everyone, so before we collect a single piece of data, we face a dilemma. We need to choose a sample size and this choice represents a crucial balancing act. On one hand, if our sample is too small, our findings will be unstable or imprecise, right? The margin of error will be large and we will have very little confidence of our conclusions.

A small sample size might tell us that a candidate has 45% of vote, but with a margin of error of ± 15 points, that's not very helpful. The results are very simply unreliable. On the other hand, the data collection is expensive.

If our sample is too large, we are wasting a lot of time, effort, resources. We might spend twice the money and time for a tiny negligible increase in precision. So this basically is a trade-off.

Our goal therefore is to find exact how many or if I have to frame it better, not exact, minimum number of sample size that will give us the precision and confidence that we are looking for. So we want to make a good decision without any unnecessary waste. That's the entire logic or the key thing behind this entire video.

How much data do we really need? So if you think of sample size, sample size is nothing but how many candidates or how many observations do you need in a sample. So let's say if you are going ahead and investigating a population about average height of people living in this country, right? So you might want to take smaller samples from various different cases in India. And those samples might have a sample size.

So let's say if you take a sample from Kerala with a sample size of 50, it means that you will go to Kerala, investigate 50 different peoples in Kerala and ask their heights and then you would have each individual telling you that their height is this, that and so on. That's what is the meaning of sample size. What determines this magic number which is the sample size? I told you that minimum sample or minimum sample size that we need to estimate the statistic that we are looking for more reliably, right? The sample size calculation is driven by three key ingredients.



Three questions you must answer before you can know how many people to survey or how many experiments to run. The very first thing is margin of error. And this margin of error, we called it as E. This is your tolerance for error.

How precise do you need your estimate to be? Do you need to know the average student score to be within plus or minus five points or plus or minus one point or whatever, right? A smaller, tighter margin of error demands a much larger sample. So if your margin of error needs to be small, in that case, you will need to have much wider sample size. And that's what would happen.

This is also very intuitive, right? If you want your estimate or the entire statistics that you are calculating, right? If you want that to be really, really close to the true value, you would need more and more samples. Or let's say what I mean is you would mean your sample size to be long enough or you would survey more number of people so that you include more variability in it. And hence, your margin of error would be smaller.

So if you would want margin of error to be small, you will have to investigate more people or you will have to include more observations in the sample. That's what it means. The second thing is confidence level.

So how certain do you want to be that your sample's confidence interval actually captures the true population parameter? Are you okay with 90% confidence or you are okay with 99% confidence? Higher confidence means you need to collect more data. This level determines the z value in our formula, right? So if you want to be 99% confidence, are you confident? Then in that case, you will require more data or more sample size. If you want to be 95% confident, then you would require slightly lesser and so on, right? So that's what it means.

The third thing is population variability. How diverse is the group you are studying, right? So let's say, for example, you are studying a population which has a lot of variability. If you have a lot of variability, in that case, you will have to take more sample size.

If the population that you're studying is not having a lot of variations, then you might be good enough with a very small sample size. So all of these three things constitute how much should be your sample size. First, the margin of error.

Second, the confidence level. Third, the population of variability. This is the entire formula.

Let's look at this formula and let's try to understand what each component means in this formula. The formula at the first look might look intimidating, but it's just a rearrangement of margin of error formula we have seen before. Okay, you would get that idea.



So this requires sample size n is equals to the quantity z score, which is our standard normal distribution. And it is also divided by the population standard deviation σ . And you have e , margin of error.

Now, margin of error is in the denominator, right? And what does that mean? If you want smaller error or smaller margin of error, maximum acceptable margin of error, in that case, your number of sample size would increase because it is indirectly proportional or also it is indirectly and is quite proportional, right? So if you want your margin of error to half, in that case, your sample size would become four times because of the squared nature and because it is inversely proportional, right? So, and if you see here, your best estimate of the population standard deviation σ is also here. Generally, we do not know that. So we use sample statistics or sample standard deviation.

That's what we know, right? Or we try to use an estimate of population standard deviation using sample standard deviation. So the key insight here in this entire thing, if you think of, is the margin of error is squared in the denominator. Having the desired error will quadruple the required sample size.

That's what I told you, right? So entire sample size calculation requires three different things. One, the level of significance at which you can constitute the Z distribution, your Z value, and then your standard deviation, population standard deviation and the margin of error. These three things, if you have, you can easily calculate what should be your effective sample size.

Now, a sharp eye student, let's say, for example, might see a paradox here, right? Some of you might still see that coming. The formula to calculate our needed sample size requires us to input σ , which is and thus the population standard deviation, which generally is unknown. We do not know about it, right? So how do we get an estimate of standard deviation before we calculate our data or before we calculate the sample size? So here are a few practical situations.

First, use a pilot study, conduct smaller preliminary samples, and then use those samples to estimate the population standard deviation using this sample standard deviation. The second thing is use data from similar studies. You can use population standard deviation or which was derived in previous studies.

Some studies might have been done previously. Let's say if you have to know average height of people living in India and you want to know how much sample size do you need for estimating that, there might have been a lot of studies which might have been done previously, right? So you'll take that standard deviation from that. Now, there are chances that the entire population might have changed over the period of time, but



when you take a look into average height of people living in India, it does not change a lot, right? People used to be 5 foot 5 and 5.5 feet on an average.

Now also, it's more or less similar. Then third, you have rule of thumb and this basically means that you can estimate likely minimum and maximum value of the population using a simple rough estimate which is range by 4. So this is a simple rough estimate that you can do, but generally we can go ahead with rule number 1 and rule number 2. Rule number 1 is most widely used because what we do is we take fewer samples and then estimate population standard deviation using sample standard deviation and then use that formula. Generally, that is what we go ahead with.

Now, what if we are not estimating the sample mean? We are trying to estimate the proportion or percentage. Let's say how many percentage of people or what proportion of the entire population has voted for a particular party or a particular candidate or whatsoever, right? The formula in that case changes like this. You have sample size, the proportion, p as proportion or percentage, okay, p into $1 - p$. This remains the same.

You have $Z_{\alpha/2}$ and n . In this particular case, if you think of, the standard deviation calculation has gone, right? So this is the entire formulation when you have to estimate for a situation wherein you are estimating a percentage. In that case, you would use sample derived percentage p . You will use that in the formula and derive the number of samples required. The term $p(1 - p)$ is maximised when p is equal to 0.5. If you think of, let's say you can also derive that by taking the derivative, putting equals to 0 and then you can find out when the p value, the maximum value of p is or the maximum value of this is when p is equal to 0.5. So that's what we generally use.

We directly use p is equal to 0.5 and then we estimate the entire thing. Now, this is an interactive explorer, a simulation based output wherein what you can do is you can change confidence level, standard deviation, margin of error. Remember that these are the three important things on which the sample size depends on.

Now, let's say that at 95% confidence interval, standard deviation 50, margin of error 5, we require 385 samples. Now, remember that in our formula, margin of error was at the bottom and squared form of it in the denominator, right? If you reduce margin of error, you should be seeing a lot of sample size required, right? This is what you see. When you go to 1, you actually require 9604 as a sample size, which is huge.

In every sample, you will have to investigate 9604 people if you are trying to estimate the average height of people living in India, which is huge. If your margin of error is 6, you will require 267. If your margin of error is 16, you will need lesser people and so on.



That's the entire intuition, right? Similarly, if you change the standard deviation, right? If you see here, if the standard deviation is large, which means the population is more variable, you would need to have more sample size. If your standard deviation is less, your population is not varying a lot, you'll need lesser sample size. If you need, let's say, let's keep this as at 70 or let's say 50.

Now, if you need to be more confident, then you would need more sample size. With less confidence level, you will need lesser sample size. That's what it means and this is all it is.

So, now in this entire lecture, you have studied about what sample size is, how do we constitute sample size, what are the three important things that a sample size depends on and then we went through the entire simulation and learned about those, how those three things impact sample sizes. So, let's meet in the next video. In summary, sample size determines and ensures that our results are both precise and cost effective.

By setting a target margin of error and confidence level, we can plan our studies more effectively. In the next lesson, we'll learn more about hypothesis testing, which will help us formally evaluate claims about populations.

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